

KUNAL KUMAVAT

Python Developer

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PROFESSIONAL SUMMARY

Python developer with 3+ years of experience in building production-grade web applications and AI-powered tools. Specialized in GenAI solutions leveraging LLMs, LangChain, RAG pipelines, and MCP architecture. Hands-on with FastAPI, Flask, and Azure Cloud services and data analysis. Passionate about technology to resolve engineering problems through AI and automation.

WORK EXPERIENCE

Associate Engineer

Mar 2023 - Present

Magna Steyr India, Pune

- Developed and maintained 5+ end-to-end web applications in Python, covering backend logic, API creation, database integration, and front-end templating, deployed to Azure Cloud services with proper access control the organization's global users.
- Built REST APIs using FastAPI and Flask to serve data between 2 client-facing applications. Implemented Pydantic for input data validation and structured LLM-generated outputs, ensuring a consistent request and response format.
- Built GenAI features involving LLMs, RAG pipelines, and LangChain—enhanced features such as document parsing and processing, semantic contextual search, and knowledge-based output generation—result in efficiency increased by 2x.
- Analyzed operational data end-to-end: queried databases, cleaned and shaped datasets, built Power BI dashboards in Power BI Desktop, and published them to Power BI Service with scheduled refreshes and row-level security — saving ~3 hours of manual reporting per week across teams.
- Identified repetitive manual processes and created automation solutions that eliminated the need for manual work, reducing time spent on recurring tasks by up to 80%.
- Managed Azure Cloud services for production-grade applications. Configured services such as Web App, Blob Storage, SQL Database, Cosmos DB, Qdrant Vector DB, and Azure AI Foundry; set up CI/CD deployment pipelines; and designed database schemas to handle semantic search and ranking across 50,000+ documents as part of a RAG pipeline.

PROJECTS

AI Conflict Detection system

- Built a document comparison engine using LangChain and Azure OpenAI GPT-4 that finds semantic conflicts between two specification documents regardless of how differently they are worded. Processes 500+ document pairs per month at ~92% detection accuracy.
- Classified conflicts into three types, such as value conflicts, implementation conflicts, and conceptual conflicts, with a short impact summary for each. Brought review time down from ~4 hours to under 30 minutes per document pair.
- Extended the system to compare flowcharts and architecture diagrams using multimodal AI, covering all specification formats used in the customer review process.
- Deployed on Azure with Entra ID access control for 50+ users; cut the overall specification review cycle by 70%.

AUTOSAR Data K-Matrix Representation Portal

- Built a web portal with FastAPI that reads ARXML files, parses them into a relational database, and serves the data through REST APIs—reduced manual data extraction from 2 days to under 10 minutes for files up to 200MB.
- Converted raw automotive communication data from CAN, LIN, and Ethernet ECU clusters into AUTOSAR K-Matrix format, replacing a spreadsheet-based process that engineers were doing by hand.
- Used by 10,000+ engineers worldwide—handles 1,000+ API requests per day at 99.5% uptime
- Setup of an ETL (extract, transform, load) pipeline to process the XML files of AUTOSAR data into a database and fetch back as per the user query and transform into the user-friendly confusion matrix table.

AI-Powered Engineering Automation Tool

- Built an AI automation tool to intelligently automate the tasks to create a bounding box on a 3D part model.
- Used the multivision LLM to analyze the CAD model dimension coordinates and route them to the correct backend REST APIs.
- The tool saves around 80% of manual efforts and completes the task in half of the time.
- Replaced the repetitive report generation task with the automated flow of presentation.

TECHNICAL SKILLS

- Languages: Python, SQL, JavaScript
- Frameworks & Tools: FastAPI, Flask, RSET API, Pydantic, Jinja2, Streamlit, Docker, Git
- Databases: SQL, NoSQL (MongoDB), vector databases (Qdrant), PostgreSQL, SQLite
- AI: LangChain, LangGraph, RAG (Retrieval-Augmented Generation), MCP (Model Context Protocol), Prompt Engineering, Ollama, OpenAI API, Multimodal AI (text + image), Semantic Search, Vector Embeddings
- Cloud services: Azure - AI Foundry, Web App, App Registrations, Entra ID, AI Search, SQL Database, Cosmos DB, Blob Storage
- Data Analysis and Visualization: DAX, Power BI Desktop, Power BI Services

AWARDS

Rising Tech Star - Rank 1 at Magna's Annual Technology Day, 2025

Awarded for the development of an AI-powered engineering automation tool. Recognized for applying AI capabilities with automation to a real, repeated engineering problem and delivering it as a working product within the organization.

EDUCATION

Bachelor of Engineering - Computer Engineering

Aug 2019 - Aug 2023

D Y Patil School of Engineering and Academy, Pune | CGPA: 8.29/10